Phosphoglycerides:

Phosphoglycerides are also mentioned as glycerolphospholipids because they contain phosphate. They are the major constituents of lipid bilayer of membrane. Phosphoglycerides have three components

(i) tri-carbon (C1, C2, C3) backbone of glycerol,

(ii) 2 long chain fatty acids esterified to C1 and C2 and

(iii) phosphoric acid esterified to C3 of glycerol.

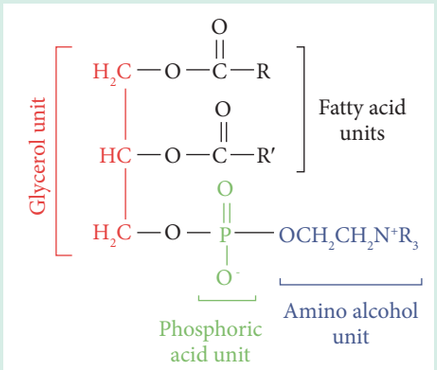


Figure 1.2 Structure of phosphoglycerides

They are amphipathic (as they have both hydrophobic (water fearing) and hydrophilic (water loving) parts. The shape of phosphoglycerides in membrane is roughly rectangular (Figure 1.2). The fatty acids which are aliphatic may have or may not have double bond or can have 2 or more double bonds.

Table 1: Common head groups found in glycerophospholipids and their basic characteristics. Source: Lafer, Eileen. "Membrane Lipids I and II: Glycerophospholipids and Sphingolipids." Lecture. San Antonio, Texas. December 6, 2011

Name of glycerophos-pholipid	Name of X	Formula of X	Net charge (at. pH7)
Phosphatidic acid	-	-H	-1
Phosphatidylethanol-amine	Ethanolamine	$-CH_2-CH_2-\overset{+}{N}H_3$	0
Phosphatidylcholine	Choline	$-CH_2-CH_2-\overset{+}{N}(CH_3)_3$	0
Phosphatidylserine	Serine	$-CH_2-\underset{COO^-}{\underset{\vert}{CH}}-\overset{+}{N}H_3$	-1
Phosphatidylglycerol	Glycerol	$-CH_2-\underset{OH}{\underset{\vert}{CH}}-CH_2-OH$	-1

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